

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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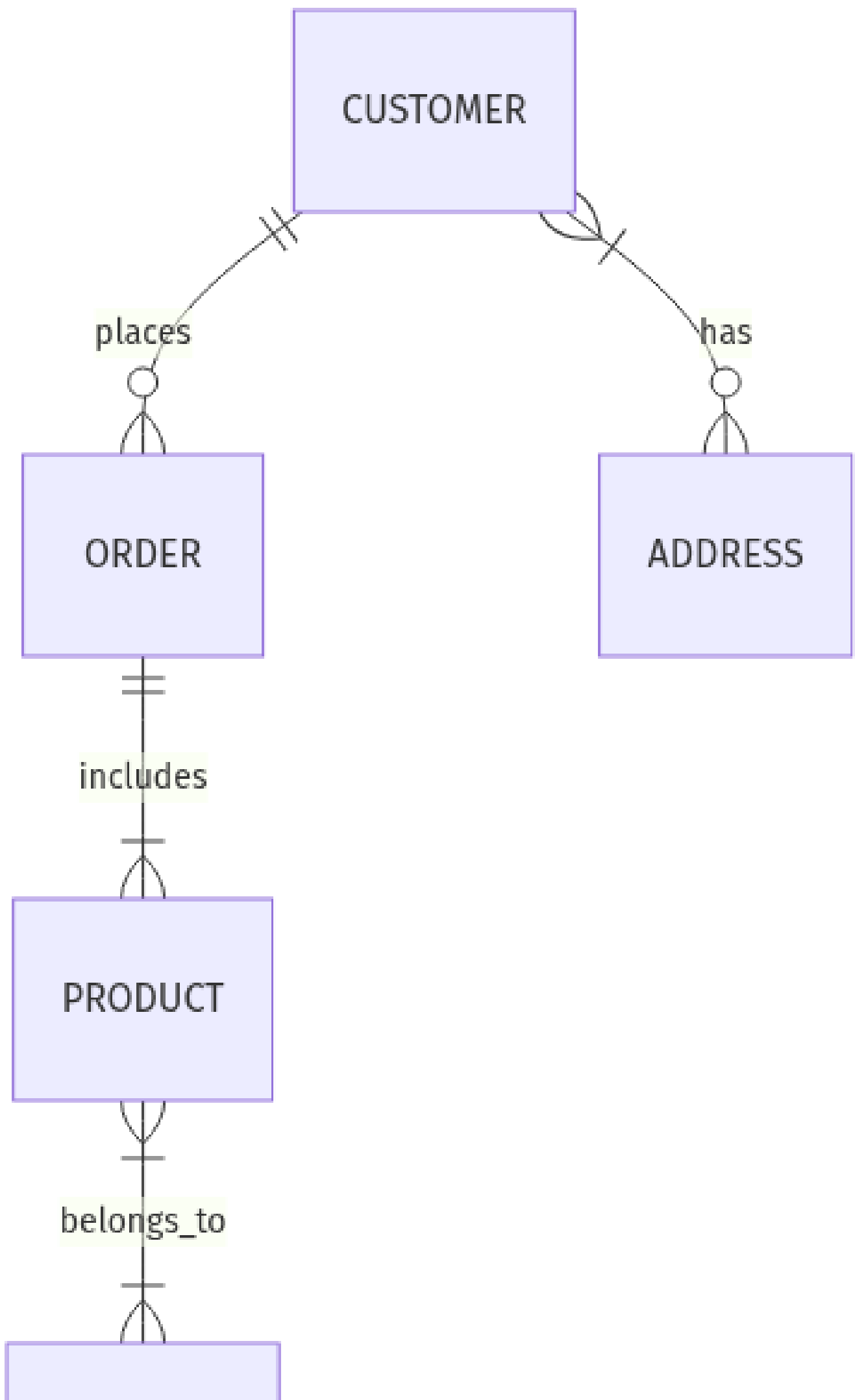
Total Marks: 0

Examination Paper

Question 1 (20 marks)

- a) Identify all the main entities and their attributes from the given scenario. (5)
- b) Determine the relationships between the identified entities and specify their cardinalities. (5)
- c) Draw the ER diagram representing the entities and their relationships detailed in the scenario. [DIAGRAM PLACEHOLDER: Draw the ER diagram for the scenario in the answer booklet.] (5)
- d) Map the identified entities and relationships to a relational schema. (5)

Figure: Diagram



Figure

```
erDiagram
    CUSTOMER ||--o{ ORDER : places
    ORDER ||--|{ PRODUCT : includes
    CUSTOMER }|--o{ ADDRESS : has
    PRODUCT }|--|{ CATEGORY : belongs_to
```

Question Q2 (15 marks)

- a) Define the key concepts and their importance related to frame bytes time. (2)
- b) Describe the main components and how they interact related to frame bytes time. (1)
- c) Analyze the advantages and disadvantages of different approaches related to frame bytes time. (2)
- d) Compare different implementation strategies and their trade-offs related to frame bytes time. (3)
- e) Evaluate the effectiveness of the proposed solution related to frame bytes time. (2)
- f) Explain the real-world applications and use cases related to frame bytes time. (2)
- g) List the key features and their benefits related to frame bytes time. (3)

Question Q3 (25 marks)

- a) Define the key concepts and their importance related to Transaction Management and Concurrency. (4)
- b) Describe the main components and how they interact related to Transaction Management and Concurrency. (2)
- c) Analyze the advantages and disadvantages of different approaches related to Transaction Management and Concurrency. (3)
- d) Compare different implementation strategies and their trade-offs related to Transaction Management and Concurrency. (5)
- e) Evaluate the effectiveness of the proposed solution related to Transaction Management and Concurrency. (5)
- f) Explain the real-world applications and use cases related to Transaction Management

and Concurrency. (3)

g) List the key features and their benefits related to Transaction Management and Concurrency. (3)

Question Q4 (40 marks)

a) Identify the main entities and their attributes. (9)

b) Draw the ER/EER diagram showing relationships and cardinalities. (5)

c) Map the ER diagram to a relational schema with primary and foreign keys. (3)

d) Explain the relationships between entities and their cardinality constraints. (5)

e) Describe the attributes of each entity and their data types. (5)

f) List the primary keys, foreign keys, and integrity constraints. (8)

g) Design the complete relational schema based on the ER diagram. (5)

Figure: ER

ENTITY



relates_to

